SCIENCE OLYMPIAD FOUNDATION

Class 8

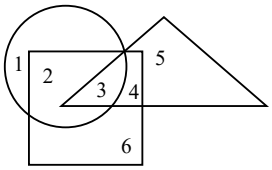
IMO

Set A

Year 2019

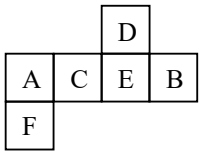
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1. In the given Venn diagram, circle represents people who live in joint families, triangle represents people who are business men and square represents married people. Which of the following number represents the married people who live in joint family but are not business man?



A. 3
B. 2
C. 4
D. 1

2. The given figure shows the net of a cube. Which of the following alphabets does not lie on the opposite faces, if the net is folded to form the cube?

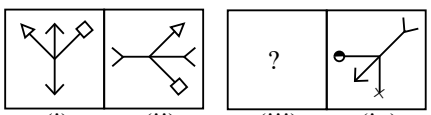


A. A and E
B. F and D
C. C and D
D. C and B

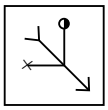
3. In an English alphabetical series, if the second half of the English alphabet series is written in reverse order, then which letter should be 8th letter to the right of 15th letter from the left end?

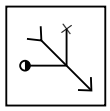
A. M
B. G
C. Q
D. Y

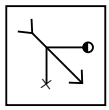
4. There is a certain relationship between figures (i) and (ii). Establish the similar relationship between figures (iii) and (iv) by selecting a suitable figure from the options which would replace the (?) in Fig. (iii).



A. 

B. 

C. 

D. 

5. Select a figure from the options which will complete the pattern in the given figure.



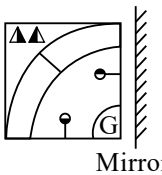
A. 

B. 

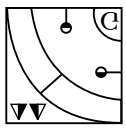
C. 

D. 

6. Select the correct mirror image of the given figure.



A. 

B. 

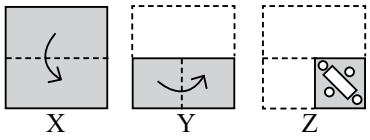
C. 

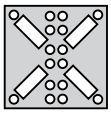
D. 

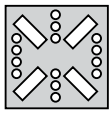
7. The given question consists of a set of three figures X, Y and Z showing a sequence of folding a piece of paper. Fig. (Z) shows the manner in which the folded

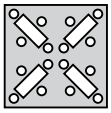
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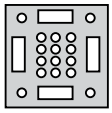
paper has been cut. Select a figure from the options which would most closely resemble the unfolded form of Fig. (Z).



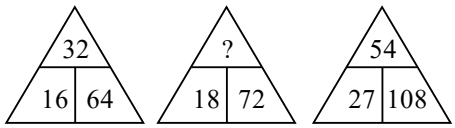
A. 

B. 

C. 

D. 

8. Find the missing number, if same rule is followed in all the three figures.



A. 42
B. 36
C. 54
D. 90

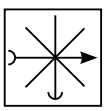
9. If 'A # B' means 'B is father of A', 'A * B' means 'B is mother of A', 'A £ B' means 'B is brother of A' and 'A $ B' means 'B is sister of A', then which of the following means 'O is grandmother of T'?

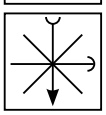
A. R£T#M#P*O
B. T£R#M$P*O
C. O*P$M#R£T
D. None of these

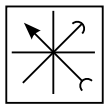
10. If it is possible to make a meaningful word with the second, fifth, seventh and eighth letters of the word VISIONARY, then which of the following will be the third letter of that word? If more than one such words can be formed, give 'Z' as the answer. If no such word can be formed, give 'X' as the answer.

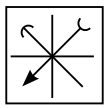
A. X
B. A
C. Z
D. R

11. Select the odd one out.

A. 

B. 

C. 

D. 

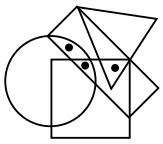
12. If '*' is written as '×', '$' is written as '+', '#' is written as '÷' and '©' is written as '–', then what is the value of 288 # 16 $ 65 * 5 © 146 ?

A. 153
B. 242
C. 197
D. 185

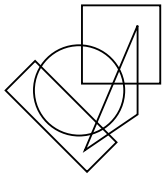
13. Persons P, Q, R and S live in red, green, yellow and blue coloured houses placed in sequence on a street. R lives in a yellow house. The green house is adjacent to the blue house. P does not live adjacent to R. The yellow house is in between the green house and red house. The colour of the P's house is _____.

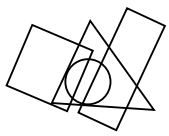
A. Blue
B. Green
C. Red
D. Data inadequate

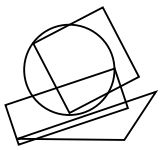
14. Which of the following options satisfies the same conditions of placement of the dots as in the given figure?



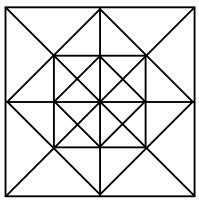
A. 

B. 

C. 

D. 

15. How many squares are formed in the given figure?



A. 23
B. 26
C. 22
D. None of these

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16. Arrange the steps of construction while constructing a quadrilateral $PQRS$ given that PQ = 4.2 cm, PS = 3 cm, QR = 1.5 cm, $\angle P = 60°$ and $\angle Q = 75°$.

Steps of construction:

Step 1 : With Q as centre and radius 1.5 cm, cut off QR = 1.5 cm along QX.

Step 2 : Construct $\angle QPY = 60°$ at P.

Step 3 : Join RS.

Step 4 : With P as centre and radius 3 cm, cut off PS = 3 cm along PY.

Step 5 : Draw PQ = 4.2 cm.

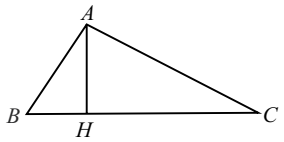
Step 6 : Construct $\angle PQX = 75°$ at Q.

A. 5, 2, 4, 1, 3, 6
B. 5, 4, 2, 1, 6, 3
C. 5, 2, 4, 6, 1, 3
D. 5, 2, 4, 1, 6, 3

17. Find the value of $\left(\sqrt{\frac{625}{4356}}+\sqrt{\frac{576}{1089}}\right)\times\left(\frac{66}{\sqrt{19600}+\sqrt{36}}\right)$.

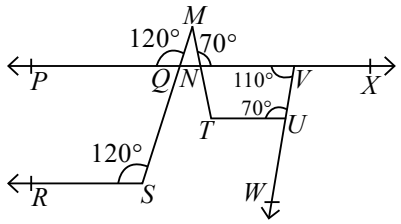
A. $\frac{7}{15}$
B. $\frac{9}{53}$
C. $\frac{1}{2}$
D. $\frac{79}{33}$

18. ABC is a right angled triangle with $\angle BAC = 90°$. AH is drawn perpendicular to BC. If AB = 60 cm and AC = 80 cm, then BH = ____.



A. 36 cm
B. 32 cm
C. 24 cm
D. 30 cm

19. If all lines are straight lines in the given figure (not drawn to scale), then which of the following options is incorrect?



A. $PQ \parallel RS$
B. $\angle NVU + \angle NTU = 180°$
C. $MT \parallel VW$
D. None of these

20. The sum of the order of rotational symmetry of Figure-P and Figure-Q is _____.

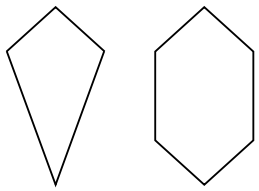


Figure-P Figure-Q

A. 0
B. 1
C. 2
D. 3

21. Find the median and mode respectively of the given data.

92, 42, 62, 86, 48, 29, 62, 64, 42, 62, 82

A. 62, 62
B. 42, 42
C. 62, 42
D. 82, 62

22. A die is thrown once. Find the probability that the number on the uppermost face is

(a) 5
(b) not a multiple of 2.

	(a)	**(b)**
A.	5/6	1/6
B.	1/2	3/2
C.	1/6	1/2
D.	3/4	1/4

23. Simplify :

$$\frac{\left(\frac{4}{3}\times\left(-\frac{25}{2}\right)\right)+\left(\left(-\frac{10}{3}\right)\times\frac{5}{2}\right)-\left(\left(-\frac{16}{3}\right)\times\left(\frac{-45}{32}\right)\right)}{\frac{3}{4}\times\left(\frac{9}{14}\times\left(-\frac{2}{18}\right)\right)}$$

A. $13\frac{11}{27}$
B. $606\frac{2}{3}$
C. $-133\frac{7}{4}$
D. $606\frac{7}{3}$

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24. Which of the following options is incorrect?

A. A quadrilateral having exactly one pair of parallel sides is called a trapezium.

B. The measure of each exterior angle of an n-sided regular polygon is $\left(\frac{180°}{n}\right)$.

C. A quadrilateral is a rhombus, if its all sides are equal and diagonals bisect each other at right angles.

D. None of these

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25. Vineet invests a sum of ₹ 41250 at the rate of 5% per annum. What compound interest will he obtain at the end of 3 years?

A. ₹ 8100.10

B. ₹ 6502.03

C. ₹ 8210.70

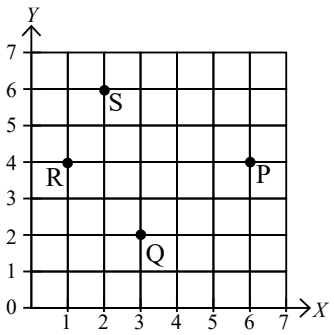
D. ₹ 7879.04

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26. Study the given graph carefully and find

(i) the coordinates of P

(ii) the difference between the abscissa of S and ordinate of Q.



	(i)	(ii)
A.	(2, 4)	2
B.	(4, 6)	1
C.	(6, 4)	0
D.	(4, 4)	0

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27. If the distance travelled by a bus in 2 hours is 310 km, then the distance travelled by the same bus with the same speed in 360 seconds is _____.

A. 20 km

B. 3 km

C. 3.5 km

D. 15.5 km

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28. Two cubes have volumes in the ratio 27 : 216. What is the ratio of the area of the face of one cube to that of the other cube?

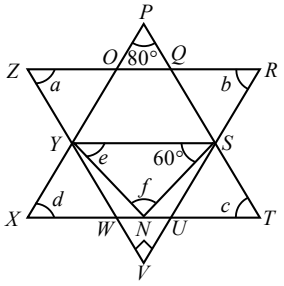
A. 1 : 4

B. 1 : 6

C. 1 : 9

D. 1 : 18

29. Find the value of $(a + b) - (e + f) + (c + d)$, if $\angle WVU = 90°$.



A. 90°

B. 70°

C. 200°

D. 210°

---

30. A 5-digit number $xy235$ is divisible by 3 such that $x + y < 5$, where x and y are single digits, then possible values of (x, y) are ____.

A. (1, 1) or (4, 0)

B. (1, 1) or (2, 0)

C. (1, 1) or (0, 2)

D. (2, 0) or (0, 2)

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31. Factorise : $\sqrt{2}y^2 + 8y + 6\sqrt{2}$

A. $\left(y + 3\sqrt{2}\right)\left(\sqrt{2}y + 2\right)$

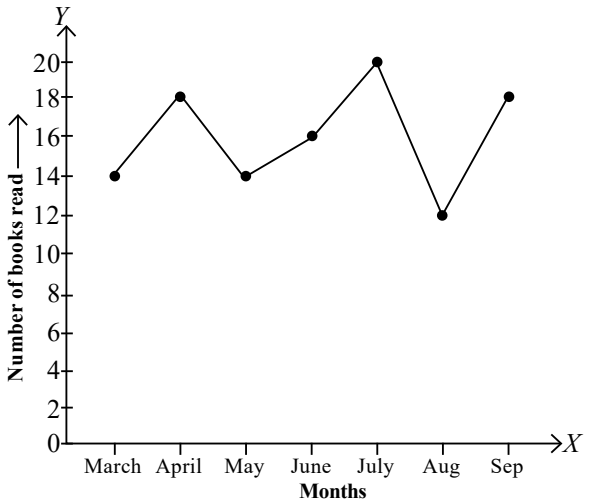
B. $\left(y + \sqrt{2}\right)\left(\sqrt{2}y + 2\right)$

C. $\left(y + 3\sqrt{6}\right)\left(\sqrt{2}y + 2\right)$

D. None of these

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32. The given line graph shows the number of books read by Shivani in 7 months.



What is the ratio of number of books read in May and July together to the total number of books read in 7 months?

A. 12 : 71

B. 3 : 7

C. 17 : 56

D. 16 : 7

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33. Solve for m : $\frac{(m+2)(2m-3)-2m^2+6}{m-4}=7$

A. $\frac{14}{3}$
B. 10
C. 15
D. $\frac{20}{3}$

34. A card is drawn from a packet of 50 cards numbered 1 to 50. Find the probability of drawing a card with a prime number.

A. 2/25
B. 7/25
C. 3/10
D. None of these

35. If a, b and c are rational numbers, then associativity of rational numbers under multiplication is given by

A. $a \times b = b \times a$
B. $a + (b + c) = (a + b) + c$
C. $a \times (b \times c) = (a \times b) \times c$
D. $a \times (b + c) = (a \times b) + (a \times c)$

EVERYDAY MATHEMATICS

36. Raghav spent ₹ 55760 on the interior decoration for his home, ₹ 37896 on buying air conditioner and the remaining 54% of the total amount he had as cash with him. What was the total amount?

A. ₹ 203600
B. ₹ 102300
C. ₹ 134560
D. ₹ 230600

37. A rectangular block of ice measures 40 cm by 25 cm by 15 cm. Find the weight of the block, if ice weighs $\frac{9}{10}$ of the weight of the same volume of water and 1 cm^3 of water weighs 1 g.

A. 9 kg
B. 13.5 kg
C. 8 kg
D. 9.5 kg

38. 4 years ago, the average age of Sakshi, Anjali and Samridhi was 32 years. 6 years ago, the average age of Anjali and Samridhi was 30 years. Find the present age of Sakshi.

A. 50 years
B. 48 years
C. 36 years
D. 42 years

39. If $(x - 5)$ notebooks cost ₹ $(x^2 - 13x + 40)$, then what is the cost of one notebook?

A. ₹ $(x - 1)$
B. ₹ $(x - 2)$
C. ₹ $(x - 6)$
D. ₹ $(x - 8)$

40. There are few males and females in a meeting. If $\frac{3}{8}$ of the people in the meeting are males and there are 90 more females than males, then how many females are there in the meeting?

A. 135
B. 200
C. 225
D. None of these

41. A bacterial cell doubles itself after every 1 hour. How many cells will there be after 8 hours?

A. 200 times of the original
B. 2^{10} times of the original
C. 2^8 times of the original
D. 2^6 times of the original

42. Manpreet took part in an interschool race competition. He cycled 2.7 km, ran for $\frac{2}{3}$ hour and walked the remaining 500 m of the race. If the total distance of the race was 12 km, then find his running speed.

A. 13.2 km/hr
B. 12 km/hr
C. 7.5 km/hr
D. 8.8 km/hr

43. 200 persons had food provision for 48 days. If 40 persons left the place, then the food will lasts for _____.

A. 60 days
B. 120 days
C. 40 days
D. 80 days

44. The outer dimensions of a closed box are 15 cm by 13 cm by 10 cm. Thickness of the wood is 2 cm. Find the total cost of wood required to make the box, if 1 cm^3 of wood costs ₹ 5.00.

A. ₹ 6780
B. ₹ 2020
C. ₹ 4880
D. ₹ 8210

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45. The results of a random survey showed that 22 out of 75 people are suffering from viral fever. Which is the best prediction of the total number of people suffering from fever, if there are 3000 people in a village?

A. 2500
B. 500
C. 900
D. 2000

ACHIEVERS SECTION

46. Read the given statements carefully and select the correct option.

Statement-1 : If the radii of two right circular cylinders are in the ratio 1 : 2 and their heights are in the ratio 4 : 3, then the ratio of their curved surface areas is 2 : 3.

Statement-2 : If each edge of a cube is doubled, then the surface area of the new cube will become three times.

A. Statement-1 is false but Statement-2 is true.
B. Statement-1 is true but Statement-2 is false.
C. Both Statement-1 and Statement-2 are true.
D. Both Statement-1 and Statement-2 are false.

47. Fill in the blanks and select the correct option.

(i) Product of a positive rational number with a __**P**__ rational number is always negative.

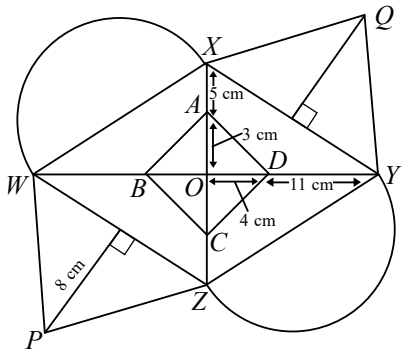
(ii) Multiplicative inverse of $\frac{1}{3^{-14}}$ is __**Q**__.

(iii) The unit digit of the square of the number 1989 is __**R**__.

(iv) If three acute angles of a quadrilateral measure 80° each, then the measure of the fourth angle is __**S**__.

	P	**Q**	**R**	**S**
A.	Positive	3^{-14}	3	140°
B.	Negative	3^{14}	3	150°
C.	Positive	3^{-14}	9	110°
D.	Negative	3^{-14}	1	120°

48. In the given figure (not drawn to scale), *ABCD* and *WXYZ* both are rhombus, 2 semi-circles and 2 identical triangles are drawn on the sides of rhombus *WXYZ*. Which of the following options is correct?



A. Area of ΔWPZ is 104 cm^2.
B. Circumference of the semi-circle is 20 cm.
C. Area of rhombus *ABCD* is 24 cm^2.
D. None of these

49. Read the given statements carefully and state 'T' for true and 'F' for false.

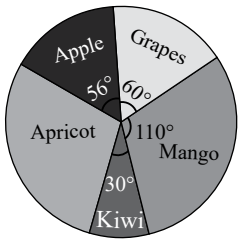
(i) The compound interest on ₹ 60000 for 3 years, if the rate of interest is 5% for first year, 4% for second year and 3% for third year is ₹ 7485.60.

(ii) If the compound interest on a sum for 2 years is ₹ 832 and the simple interest on the same sum at the same rate for the same time period is ₹ 800, then the rate of interest is 12%.

(iii) For a certain sum, if the simple interest for $3\frac{1}{2}$ years at 8% p.a. is ₹ 60 more than the simple interest for $2\frac{1}{2}$ years at 10% p.a. for the same sum, then the sum is ₹ 2000.

	(i)	**(ii)**	**(iii)**
A.	T	T	F
B.	F	T	F
C.	T	F	T
D.	F	F	T

50. The given pie chart shows the different types of fruits in a store. The total number of fruits in the store is 1080.



If a fruit is selected at random, then

(a) Find the probability that the selected fruit is an Apricot.

(b) Find the probability that the selected fruit is a Kiwi.

	(a)	**(b)**
A.	$\frac{19}{60}$	$\frac{29}{60}$
B.	$\frac{17}{45}$	$\frac{7}{22}$
C.	$\frac{13}{45}$	$\frac{1}{12}$
D.	$\frac{23}{50}$	$\frac{17}{40}$

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